

JITHU S PILLAI

Software Engineer

+9162353-68897

jithu.codes@gmail.com

LinkedIn

GitHub

LeetCode

Profile Summary

Results-driven Software Engineer with deep expertise in the MERN stack, Generative AI, and automation. Passionate about building scalable web applications and solving real-world challenges through clean, maintainable code. Strong understanding of modern software architectures, design patterns, and AI integrations, with a keen interest in contributing to open-source projects and the developer community.

Technical Skills

Languages: JavaScript, TypeScript

Frontend: HTML, CSS, SCSS, React.js, Next.js, Redux, Redux Toolkit (RTK), Tailwind CSS, Bootstrap, Framer Motion, EJS

Backend & APIs: Node.js, Express.js, Fastify, RESTful APIs, Real-time WebSockets, JWT, OAuth 2.0 (Google), Razorpay, Passport.js

Databases: PostgreSQL, MongoDB, Redis, Supabase, Pinecone

AI & Automation: LangChain.js, RAG, Hugging Face, Gemini AI, n8n

DevOps & Cloud: Docker, AWS (EC2, S3, CloudFront), CI/CD, Nginx, Cloudinary, Render, Hostinger, Firebase, Linode, Cloudflare

Architecture & Concepts: System Design, DB Design, Clean Architecture, Layered Architecture, Monorepos, OOP, SOLID, MVC, SEO

Developer Tools: GitHub, Jira, Postman, Insomnia, Buf, Figma, Eraser.io, Stitch, Antigravity, Lovable, Bolt.new, Claude AI, Canva

Professional Experience

Software Engineer

Ducbot Innovations Pvt. Ltd.

Aug 2025 – Present

Remote

- Architected and developed scalable full-stack applications using React.js, Node.js, Express.js, MongoDB, and PostgreSQL across Freight & Logistics, FinTech, FoodTech, and E-Commerce domains.
- Designed and implemented secure RESTful APIs, authentication workflows, role-based access control (RBAC), and business-critical backend services.
- Led development of CRM platforms, Admin Portals, Vendor Portals, and analytics dashboards, improving operational efficiency and business workflow automation.
- Managed cloud deployment, CI/CD workflows, and production environments using AWS, Google Cloud, Firebase, Cloudflare, Linode, and Hostinger.
- Collaborated with stakeholders to translate business requirements into scalable software solutions while conducting code reviews and maintaining engineering best practices.

Associate Software Engineer

Ducbot Innovations Pvt. Ltd.

May 2025 – Aug 2025

Remote

- Developed and maintained dynamic web applications and business management systems using the MERN stack, delivering responsive and user-centric solutions.
- Contributed to the development of Temple Billing Software, implementing features for transaction management, reporting, and operational automation.
- Built reusable frontend components, integrated third-party APIs, and optimized application performance to improve scalability and maintainability.
- Designed database schemas, aggregation pipelines, and optimized queries using MongoDB and PostgreSQL to enhance system performance.
- Worked closely with senior developers in Agile development cycles, participating in sprint planning, debugging, testing, and Git-based collaboration.

Projects

MedBloom – Intelligent Telemedicine & EHR Platform

[GitHub](#)

React.js, Node.js, Express.js, MongoDB, Redis, Socket.IO, WebRTC, Gemini AI, node-cron, node-Crypto

- **AI Engineering:** Architected a **Gemini AI** patient triage engine with **strict JSON schema enforcement**, automating real-time specialty routing and emergency classifications.
- **High-Throughput Data Processing:** Engineered a **node-cron** scheduling framework driving optimized **MongoDB bulkWrite operations** and **Socket.IO triggers** to eliminate data racing and guarantee 5-minute pre-call alerts.
- **Backend Robustness & Security:** Hardened authentication against timing attacks using **crypto.timingSafeEqual**, backed by **SHA-256 token hashing** and automated **Redis TTL cache eviction**.
- **Real-time Systems:** Designed a low-latency **WebRTC signaling infrastructure** utilizing **Socket.IO dynamic namespaces** and query-routing to seamlessly stream simultaneous, high-throughput audio/video consultations.

MomentumOS - Emotionally Intelligent Productivity Engine

[GitHub](#)

React 18, TypeScript, Tailwind CSS, Zustand, Framer Motion, Node.js, Express, MongoDB, Gemini 3.5 Flash, Web Speech API

- **AI & Ingestion Pipelines:** Engineered a resilient voice-to-text task extraction pipeline utilizing the **Web Speech API** and **Gemini 3.5 Flash**, implementing **custom linear backoff retry logic** to gracefully handle transient network errors while parsing unstructured user inputs into strict, validated JSON schemas.
- **Algorithmic Optimization:** Developed a multi-variable priority scoring engine executing a multi-factor weight formula (urgency, importance, emotion) to optimize task queues prior to **MongoDB persistence**.
- **State Architecture & UI Polish:** Orchestrated a glassmorphic interface using **Zustand global state** and **Framer Motion**, synchronizing complex micro-interactions, canvas SVG gauges, and animations with real-time backend data.
- **Fault-Tolerant System Design:** Architected an automated recovery system that intercepts task failure states, triggering AI micro-steps to mitigate burnout while managing RPG-style state engines and badge lifecycles.

NaviSphere – AI Offline Indoor Navigation PWA

[GitHub](#) — [Live Link](#)

React, Node.js, Express, Redux Toolkit, Vite PWA (Workbox), Gemini 2.5 Flash, Cloudinary, Tailwind CSS, Dijkstra/A Algorithms*

- **Offline-First Architecture:** Engineered an offline-first indoor navigation platform ensuring 100% availability in network dead zones by deploying a robust **Progressive Web App (PWA)** architecture leveraging **Workbox service workers** and a **cache-first runtime strategy** for static assets.
- **Algorithmic Optimization:** Architected a custom routing engine utilizing **Dijkstra's and A* algorithms** with a **priority queue over dynamic adjacency lists**, delivering sub-second shortest-path computations.
- **Semantic Processing:** Integrated **Gemini AI** using **token-optimized context filtering** and structured prompts to map abstract natural language strings directly to physical points of interest.
- **State Architecture:** Engineered a high-frame-rate responsive mapping engine using **Redux Toolkit** to seamlessly synchronize complex, real-time client UI states across iOS/Android.

Book My Venue – Full-Stack Venue Booking Platform (In Development) Team Lead & Architect

[GitHub](#)

React.js, Node.js, Express.js, MongoDB, Redis, Socket.IO, JWT, Figma

- **Engineering Leadership:** Spearheaded a team of developers from conceptualization to execution, owning the product roadmap, database schema design, and UI/UX design wireframes within Figma.
- **Advanced Authentication:** Engineered a highly secure, dual-token authentication system leveraging **JWT access tokens** and **Redis-backed refresh tokens** for instant session revocation and stateless user verification.
- **System Architecture & Workflows:** Designed the core end-to-end booking lifecycle, vendor management portals, and multi-role access control (RBAC) to handle high-concurrency reservation workflows.
- **DevOps & Collaboration:** Established the project's Git branching strategy, led comprehensive code reviews, and managed feature integration pipelines to ensure high code quality and zero deployment bottlenecks.

Education

MERN Stack Development

2025 — Present

Brototype

B.Tech - Chemical Engineering

2018–2022

A. P. J. Abdul Kalam Technological University